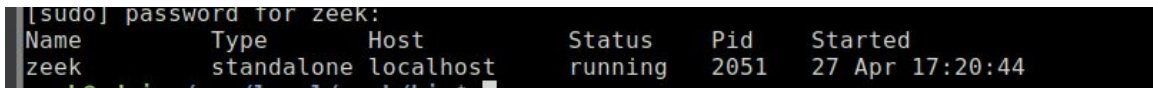


Лабораторная работа №6

Знакомство со сценариями системы Zeek

1.1 Запуск Zeek

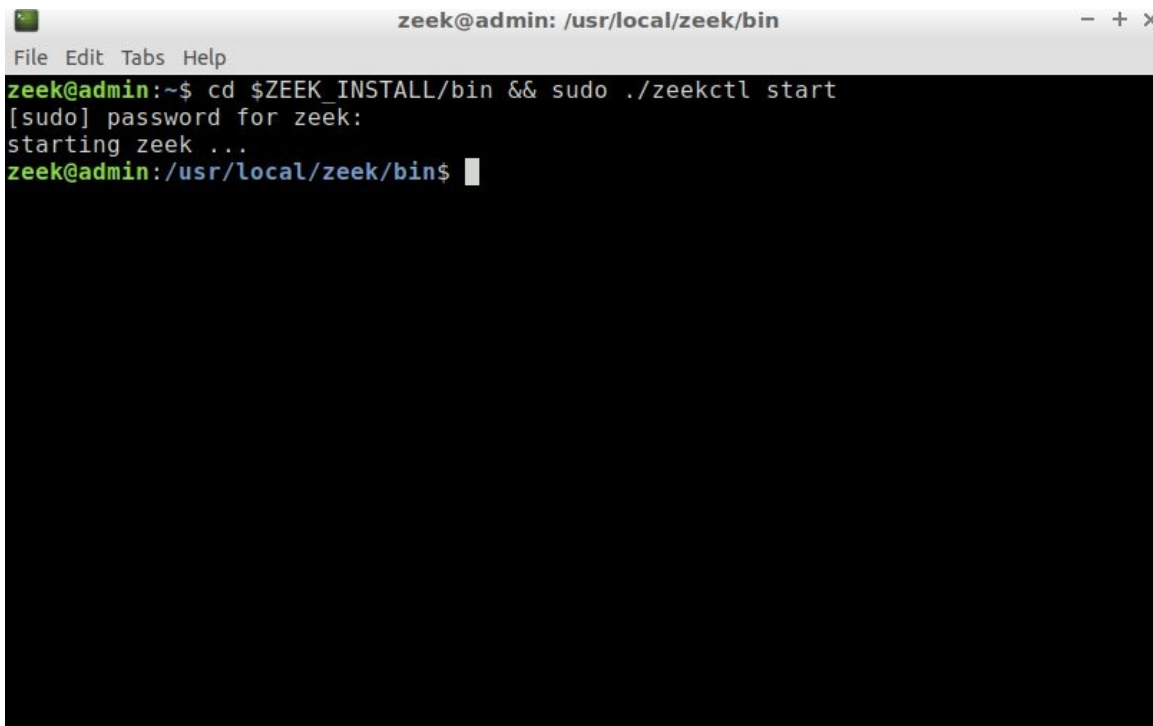
Zeek успешно запущен. Статус: running.

A terminal window showing the output of the 'sudo systemctl status zeek' command. It displays a table with columns: Name, Type, Host, Status, Pid, and Started. The row for 'zeek' shows it is 'running' with PID 2051, started on 27 Apr 17:20:44.

Name	Type	Host	Status	Pid	Started
zeek	standalone	localhost	running	2051	27 Apr 17:20:44

Рисунок 1 – Статус Zeek: running

Запуск нового экземпляра Zeek через zeekctl start.

A terminal window titled 'zeek@admin: /usr/local/zeek/bin' showing the execution of 'zeekctl start'. The prompt is 'zeek@admin:~\$'. The command executed is 'cd \$ZEEK_INSTALL/bin && sudo ./zeekctl start'. The output shows the password prompt, 'starting zeek ...', and the final prompt 'zeek@admin:/usr/local/zeek/bin\$' with a cursor.

```
zeek@admin:~$ cd $ZEEK_INSTALL/bin && sudo ./zeekctl start
[sudo] password for zeek:
starting zeek ...
zeek@admin:/usr/local/zeek/bin$
```

Рисунок 2 – Успешный запуск Zeek

1.2 Запуск UDP сценария Zeek

Переход в директорию Lab-Scripts. Просмотр содержимого сценария lab6_sec2-2.zeek.

```
zeek@admin: ~/Zeek-Labs/Lab-Scripts
File Edit Tabs Help
zeek@admin:~$ cd $ZEEK_INSTALL/bin && sudo ./zeekctl start
[sudo] password for zeek:
starting zeek ...
creating crash report for previously crashed nodes: zeek
Error: error occurred while trying to send mail: send-mail: /usr/sbin/sendmail not found

zeek@admin:/usr/local/zeek/bin$ cd ~/Zeek-Labs/Lab-Scripts/
zeek@admin:~/Zeek-Labs/Lab-Scripts$ nl lab6_sec2-2.zeek
 1 event udp_request(u: connection){
 2     print fmt("A UDP Request was found: %s", u$id$resp_h);
 3 }

 4 event udp_reply(u: connection){
 5     print fmt("A UDP Reply was found: %s", u$id$resp_h);
 6 }
zeek@admin:~/Zeek-Labs/Lab-Scripts$
```

Рисунок 3 – Содержимое сценария lab6_sec2-2.zeek

Обработка файла захвата smallFlows.pcap с использованием сценария lab6_sec2-2.zeek.

```
zeek@admin:~/Zeek-Labs/Lab-Scripts$ cd ~/Zeek-Labs/UDP-Traffic/
zeek@admin:~/Zeek-Labs/UDP-Traffic$ zeek -C -r ../Sample-PCAP/smallFlows.pcap ../Lab-Scripts/lab6_sec2-2.zeek
A UDP Request was found: 239.255.255.250
A UDP Request was found: 239.255.255.250
A UDP Request was found: 239.255.255.250
A UDP Request was found: 239.255.255.250
A UDP Request was found: 239.255.255.250
A UDP Request was found: 239.255.255.250
A UDP Request was found: 255.255.255.255
A UDP Request was found: 224.0.0.252
A UDP Request was found: 224.0.0.252
A UDP Request was found: 71.224.25.112
A UDP Request was found: 255.255.255.255
A UDP Reply was found: 172.16.0.1
A UDP Request was found: 224.0.0.252
A UDP Reply was found: 71.224.25.112
A UDP Request was found: 71.224.25.112
A UDP Request was found: 224.0.0.252
A UDP Reply was found: 71.224.25.112
A UDP Request was found: 156.26.54.15
A UDP Reply was found: 156.26.54.15
A UDP Request was found: 178.144.253.171
A UDP Request was found: 172.16.255.255
A UDP Reply was found: 178.144.253.171
A UDP Request was found: 147.31.122.1
A UDP Reply was found: 147.31.122.1
A UDP Request was found: 75.76.39.18
A UDP Reply was found: 75.76.39.18
```

Рисунок 4 – Результат выполнения сценария lab6_sec2-2.zeek

1.3 Запуск TCP сценария Zeek

Просмотр содержимого сценария lab6_sec2-3.zeek.

```
zeek@admin:~/Zeek-Labs/UDP-Traffic$ nl ../Lab-Scripts/lab6_sec2-3.zeek
 1 event tcp_packet(c: connection, is_orig: bool, flags: string, seq: count, ack: count, len: count, payload: string) {
 2     print fmt("Destination Port #: %s", c$id$resp_p);
 3 }
```

Рисунок 5 – Содержимое сценария lab6_sec2-3.zeek

Обработка файла захвата smallFlows.pcap с использованием сценария lab6_sec2-3.zeek.

Проверка содержимого директории. Файл UpdatedConn.log создан.

```
zeek@admin:~/Zeek-Labs/UDP-Traffic$ zeek -C -r ../Sample-PCAP/smallFlows.pcap ../Lab-Scripts/lab6_sec3-1.zeek
zeek@admin:~/Zeek-Labs/UDP-Traffic$ ls
conn-http.log  dns.log  files.log  packet_filter.log  ssl.log  UpdatedConn.log  x509.log
dhcp.log      dpd.log  http.log  snmp.log          udptraffic.pcap  weird.log
zeek@admin:~/Zeek-Labs/UDP-Traffic$
```

Рисунок 9 – Файл UpdatedConn.log в директории UDP-Traffic

2.2 Обновление потока conn.log

Очистка рабочей директории.

```
zeek@admin:~/Zeek-Labs/UDP-Traffic$ ../../Lab-Scripts/lab_clean.sh
zeek@admin:~/Zeek-Labs/UDP-Traffic$
```

Рисунок 10 – Очистка директории

Просмотр содержимого сценария lab6_sec3-2.zeek.

```
zeek@admin:~/Zeek-Labs/UDP-Traffic$ ../../Lab-Scripts/lab_clean.sh
zeek@admin:~/Zeek-Labs/UDP-Traffic$ nl ../Lab-Scripts/lab6_sec3-2.zeek
 1 function http_only(rec: Conn::Info) : bool {
 2
 3     return rec?$service && rec$service == "http";
 4 }
 5
 6 event zeek_init(){
 7     local filter: Log::Filter = [$name="http-only", $path="conn-http", $pred=http_only];
 8     Log::add_filter(Conn::LOG, filter);
 9 }
zeek@admin:~/Zeek-Labs/UDP-Traffic$
```

Рисунок 11 – Содержимое сценария lab6_sec3-2.zeek

Запуск сценария lab6_sec3-2.zeek на файле захвата.

```
zeek@admin:~/Zeek-Labs/UDP-Traffic$ zeek -C -r ../Sample-PCAP/smallFlows.pcap ../Lab-Scripts/lab6_sec3-2.zeek
zeek@admin:~/Zeek-Labs/UDP-Traffic$
```

Рисунок 12 – Запуск сценария lab6_sec3-2.zeek

Проверка содержимого директории. Файлы conn.log и conn-http.log созданы.

```

zeek@admin:~/Zeek-Labs/UDP-Traffic$ ../../Lab-Scripts/lab_clean.sh
zeek@admin:~/Zeek-Labs/UDP-Traffic$ nl ../Lab-Scripts/lab6_sec3-2.zeek
1  function http_only(rec: Conn::Info) : bool {
2
3      return rec?$service && rec$service == "http";
4  }
5
6  event zeek_init(){
7      local filter: Log::Filter = [$name="http-only", $path="conn-http", $pred=http_only];
8      Log::add_filter(Conn::LOG, filter);
9  }
zeek@admin:~/Zeek-Labs/UDP-Traffic$ zeek -C -r ../Sample-PCAP/smallFlows.pcap ../Lab-Scripts/lab6_sec3-2.zeek
zeek@admin:~/Zeek-Labs/UDP-Traffic$ ls
conn-http.log  dhcp.log  dpd.log  http.log  snmp.log  udptraffic.pcap  weird.log
conn.log       dns.log  files.log  packet_filter.log  ssl.log  UpdatedConn.log  x509.log
zeek@admin:~/Zeek-Labs/UDP-Traffic$

```

Рисунок 13 – Файлы conn.log и conn-http.log в директории UDP-Traffic

Лабораторная работа завершена, zeek остановлен с помощью команды

```

zeek@admin:~/Zeek-Labs/UDP-Traffic$ cd /usr/local/zeek/bin && sudo ./zeekctl stop
stopping zeek ...
zeek@admin:~/usr/local/zeek/bin$

```

Рисунок 14 – завершение работы zeek